MAHMOOD UL ISLAM

Software Engineer | Cloud & AI Specialist

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Areas of expertise

Programming Languages: JavaScript, Python
Frontend Development: Frameworks & Libraries: React.js, Vue.js, React Native, Swift (iOS), Next.js
Backend Development
Server-Side Development: Node.js, Express.js, Next.js (API & Middleware)
Serverless & Cloud-Native Serverless & Cloud-Native Backend: AWS Lambda, AWS Amplify Backend
API Development: RESTful APIs, GraphQL, Apollo GraphQL
Real-Time Communication: WebSockets, Server-Sent Events (SSE), Amazon Chime SDK, WebRTC, Socket.IO
Database Management & Storage
Relational Databases: PostgreSQL, MySQL, Amazon RDS
NoSQL Databases: Firebase Firestore, MongoDB, Amazon DynamoDB
ORM & Cloud Storage: Prisma, Sequelize, Amazon S3
Cloud & DevOps (AWS)
Compute & Containers: Fargate (ECS), EC2
Networking & API Management: API Gateway, VPC, ALB, NLB, Amazon EventBridge
Messaging & Streaming Services: SNS, SQS, SES, Pinpoint
Security & Identity Management: Cognito, IAM, Secrets Manager
Monitoring & Automation: CloudWatch, CloudTrail, Step Functions
Infrastructure as Code: CloudFormation, AWS CDK
Build & Deployment Tools
Version Control & CI/CD: GitHub, Bitbucket, GitHub Action, AWS Amplify CI/CD, CodePipeline, CodeBuild, CodeDeploy, Docker, Docker Compose
Testing & Code Quality: Testing & Automation: Jest, Puppeteer, Custom Node.js Scripts
Authentication & Security: OAuth 2.0, OpenID Connect (OIDC), AWS Amplify Auth, Auth0
AI, Machine Learning & Data Engineering
ML Frameworks & Training: Amazon SageMaker AI (training, fine-tuning, deployment), model evaluation, hyperparameter optimization, bias detection
Generative AI & Agents: Amazon Bedrock (Foundation Models, Agents, Knowledge Bases, Guardrails), prompt engineering, RAG architectures, multi-agent orchestration
Computer Vision: Custom-trained SageMaker models, semantic segmentation, image classification, Rekognition (built-in & custom labels)
NLP & Medical AI: Comprehend, Comprehend Medical, Transcribe, Polly, HealthLake
Data Pipelines & ETL: AWS Glue, Glue DataBrew, Data Firehose, Kinesis, EMR
Data Analytics & Warehousing: Athena, Redshift, QuickSuite, Lake Formation
Data Preparation & Analysis: Pandas, NumPy, Scikit-learn, data wrangling, feature selection
MLOps & Model Management: SageMaker Pipelines, Model Monitor, SageMaker Feature Store, SageMaker Processing, A/B testing, Shadow testing, Canary deployment, model registry
ML Concepts: Supervised & unsupervised learning, deep learning (CNNs, transformers, ViTs), transfer learning, ensemble methods, time-series forecasting, anomaly detection

Experience

Virtual Health Hub

Senior Software Engineer | Cloud & AI Specialist — *March 2025* **– Present**
Software Developer — ***August 2024*** *– February 2025*
Location: Saskatoon, Saskatchewan, Canada

Project: "LifeLine" (https://virtualhealthhub.ca), (https://research-groups.usask.ca/remote-presence), (https://lifelineapp.ca)
Virtual Mental Health Treatment Platform

Technologies: JavaScript, React.js, Vue.js, Quasar, React Native, Node.js, Express.js, REST APIs, PostgreSQL, SASS, CSS
AWS Services: Amplify, API Gateway, Bedrock, ECS (Fargate), RDS, EventBridge, Chime SDK, Lambda, Step Functions, S3, CloudWatch, Amazon Pinpoint, Cognito, Amazon Rekognition, Amazon Comprehend, Amazon Polly, Amazon Transcribe, Amazon Data Firehose, Athena, AWS Glue, Amazon Quick Suite.

Key Responsibilities & Contributions:
Project Leadership: Managing the Lifeline Project end-to-end, including development, cloud infrastructure, and solution architecture, ensuring scalability and compliance with healthcare standards
Full-Stack System Development:
• Built a highly scalable system using Next.js, Vue.js, React.js, Quasar, and React Native for enterprise-level healthcare applications, maintaining HIPAA compliance
• Built a fully dynamic appointment booking and medication management system for clinicians and patients, including useful analytics and notifications
• Created a multi-modal AI chatbot to generate tailored responses according to the patient's data, create session summaries & synchronize with the clinician portal; all within a secure VPC with AWS cloud infrastructure. Created a natural conversation with the AI chatbot using Amazon Polly and Amazon Transcribe
• Developed a video conference ambient listening to summarize the meeting with Chime SDK, Transcribe, S3 and Bedrock
• Built an Agentic AI system by Amazon Bedrock with the collaboration of multiple agents (AI chatbot agent, Video conference summary agent, Agent for risk analytics/alert)
• Developed and maintained APIs & WebSocket using Node.js and Express.js, backed by PostgreSQL, Prisma, Socket I/O, API gateway, ALB/NLB, ECS-Fargate, Cognito, EventBridge, SES and many more
• Created custom libraries and internal applications to enhance system efficiency and improve client management
• Developed a native custom module in Expo using Swift & Kotlin to bridge Amazon Chime SDK with React Native, enabling seamless audio/video conferencing capabilities
• Created a Vite compilation bundle to integrate Vue.js and React.js components within the same Quasar app, allowing smooth execution of Amazon Chime SDK React components alongside Vue.js
Cloud Infrastructure & Security:
• Architected and managed AWS cloud infrastructure, including Cognito, ECS (Fargate), RDS, EventBridge, Pinpoint, API Gateway, and Chime SDK for security and scale
• Implemented secure authentication mechanisms with AWS Cognito and Auth0 for seamless user management
AI & Machine Learning Integration:
• Developed an AI Agent using AWS Bedrock and Kinesis to enhance medical data processing, automation, and real-time insights
• Integrated Amazon Chime SDK with Amazon Transcribe, Rekognition, and Bedrock for real-time audio/video analysis
Feature Development for Virtual Healthcare:
• Designed and developed an appointment booking system with a complete workflow for a Virtual Clinic, optimizing doctor-patient interactions
• Developed a medication management system and an AI chatbot that integrates with the project's AI Agent, enhancing patient adherence and physician oversight
Stakeholder & Research Collaboration:
• Collaborating with stakeholders and medical professionals to enhance system efficiency and gather requirements
• Independently designed and conducted a dual-dataset validation study evaluating agentic AI safety in mental health, authoring a peer-reviewed manuscript: "Safety Evaluation of an Agentic AI Mental Health System Featuring Autonomous Reasoning, Context-Aware Personalization, and Retrieval-Augmented Generation: A Dual-Dataset Validation Study."

Project: "SkinScan" (https://virtualhealthhub.ca), (https://research-groups.usask.ca/remote-presence), (https://skinscanapp.ca)
AI wound-care Imaging Platform (End-to-End)

Technologies: Swift (iOS native), Next.js (doctor portal, admin portal), AprilTags (fiducial markers) Node.js, Express.js, REST APIs, PostgreSQL, Prisma, SSE
AWS Services: Amazon SageMaker AI, Bedrock, Rekognition, Comprehend Medical, S3, API Gateway, NLB, ALB, ECS Fargate, RDS, SNS, EventBridge, CloudWatch, Cognito

Key Responsibilities & Contributions

Project leadership & architecture: Designed the full cloud/application architecture (VPC, networking, CI/CD), aligning with healthcare privacy best practices
Computer vision & modeling:
• Built a unified DINOv3 ViT-L/16 model (PyTorch, HuggingFace Transformers) for simultaneous 10-class wound classification and pixel-level segmentation in a single forward pass, using two-phase transfer learning (head-only, then partial backbone unfreeze) with class-balanced cross-entropy, Mixup/CutMix augmentation, and native 4K resolution support via RoPE positional encoding
• Implemented a multi-stage pipeline: Rekognition gate for wound detection, the unified model on scale-to-zero async SageMaker endpoints with RLE-encoded mask generation, and AprilTag-based calibration for pixel-to-mm wound measurement
• Integrated three Amazon Bedrock agents for automated wound staging, healing trajectory assessment, and clinical recommendations generation
Native iOS app (Swift):
• Built AR-guided photo capture and library import with on-device checks; integrated Rekognition pre-screen calls to block low-quality/irrelevant photos before upload
• Secure media handling (presigned S3 uploads, retries, background transfers) and UX tuned for diagnostically useful images
• Integrated iOS push notifications via SNS/APNs for real-time case update alerts
Doctor portal (Next.js):
• Developed a clinician portal for patients, cases, waitlist, and triage
• Case view with image viewer, model prediction + confidence, status transitions, and notes/audit history
• Built waiting list management with case acceptance/rejection workflow and treatment plan updates
• Implemented real-time notifications for new case assignments
Backend (centralized, Node/Express):
• Built a centralized service on ECS Fargate that orchestrates ingestion, pre-screening, model inference, and persistence; secured with Amazon Cognito authentication, role-based access control (physician, nurse, wound specialist), rate limiting, and structured logging; productionized with a GitHub Actions CI/CD pipeline (Docker build, ECR push, ECS deploy, Prisma migrations)
• Implemented real-time SSE streaming with connection pooling, heartbeat monitoring, concurrent analysis limiting, and graceful degradation for long-running AI analysis
• Built visit versioning system tracking full case history across initial, follow-up, and discharge visits, with treatment plan versioning and doctor audit trails
• Integrated AWS Comprehend Medical for ICD-10-CM inference, RxNorm medication extraction, and PHI detection
• Implemented multi-facility administration, push notifications (SNS/APNs), and measurement screenshot management with PostgreSQL persistence via Prisma

Saaska Software Inc.

Software Engineer **April 2023 – June 2024**
Location: Saskatoon, Saskatchewan, Canada

Project: "esiKidz" (https://esikidz.com) childcare management software.
Project's usage: JavaScript, React.js, React Native, Node.js, REST APIs, MySQL, SASS, CSS, Git, Bitbucket, Trello, Jira
• SAAS development by React.js & React Native at the enterprise level
• learned data engineering by creating apps for web crawling, data extraction, and manipulation
• Experienced AWS cloud infrastructure maintenance, including Lambda, Amazon S3, API gateway
• Developed multiple internal applications & custom libraries to support the software system in efficiently managing clients
Project's specific works:
• Worked collaboratively in a team to maintain the overall architecture and develop a highly scalable system integrated with ReactJS & React-Native APPs, communicating via Node JS API services
• Performed optimization of the developer toolchain to support instant provisioning of new services and infrastructure, fully automating the deployment, and minimizing development friction
• Performed software testing, improving existing systems, and adding new features and maintained documentation on test and development operations
• Designed and implemented to improve the software's reliability, scalability, performance, and security

SELISE Digital Platforms

Software Engineer **June 2022 – Aug 2022**

Project: "Sunrise-club" (https://www.sunrise.ch/en/moments), an Event booking site for the Telco (Sunrise) users of Switzerland
Project's usage: JavaScript, TypeScript, React.js, Next.js, MongoDB, REST APIs, Storyblok CMS, Material UI, Ant Design UI, React-bootstrap, SASS, Bootstrap, CSS, GitHub, Jira
• Learned Next.js at the enterprise level, the pros and cons of using Next.js to maintain URL redirection
• Created a website with high security and maintained a web firewall
• Learned to implement OAuth 2 + OIDC
• Practiced using SWR with Axios, Crypto.js, and Lodash
• Proficient in using loggers in the project, built API in Next.js, implemented Storyblok CMS, and MongoDB database
Project's specific works:
• Maintained URL redirect with WAF (Web Application Firewall) in the production site. Creating an API & middleware through Next.js
• Maintained 3 test environments and helped my peer with fixing up the commit history on the main branch in our project
• Created documentation with code standards of the codebase of our project, implementing CSS modules into our main project
• Constructed a complete language translation system

Quantic Dynamics Ltd.

Web Developer **Feb 2019 – May 2022**
Project's usage: JavaScript, React.js, Next.js, MongoDB, Firebase, SASS, CSS, GitHub
• Built fully responsive web applications using React.js, Next.js, and Node.js with MongoDB and Firebase across mobile, tablet, and desktop views
• Architected frontend and backend solutions with RESTful API integration and asynchronous client-server communication
• Implemented data manipulation, search, and sort functionalities using Node.js

Education

• Applied Certificate in Website Design and Development: Saskatchewan Polytechnic
• Bachelor of Science in Textile Engineering: Ahsanullah University of Science & Technology (WES Canadian equivalency 4-year bachelor, Ref# 4848976)

Certifications

• **AWS Certified Machine Learning Engineer – Associate** Amazon Web Services (AWS)
Validation Number: 778b21fc058d42a39b13860d3b4afec8 (https://aws.amazon.com/verification)
Verify: https://www.credly.com/badges/0e1a5b1c-f08c-483a-bf10-23ccdba8a23b/public_url